

Columns and dataset merging

ISO: Country Code

Year: Year

GovIndex(%): So basically Government Index has 6 different ways of being measured(indicators). I removed that column, then GovIndex is the core governance score.

Definition:

- A standardised score ranging roughly from -2.5 (very weak) to $+2.5$ (very strong) governance performance.
- It is a z-score (standard normal unit).

I added 2.5 to the whole column, divided by 5, hence made it into a ratio, then multiply by 100 to make into a percentage

percentile rank: If you made a normal distribution of all countries and there GovIndex percentages in a single year this rank tells you how much on the right will this country be(basically how better is it than other countries in the same year on the GovIndex scale)

GovIndex_ErrorPercent: Took the stddev(Standard Deviation/ error of the GovIndex) divided it by 2.5 to make a ratio, multiplied by 100, and that's how I get the percentage

GDP per capita growth (annual %): GDP per Capita growth

HDI_percent: HDI is a composite index created by the United Nations Development Programme (UNDP)

1. Health
2. . Education
3. 3. Standard of Living

Each of the three dimensions is normalised into a **0 → 1 scale**, and then:

$$\text{HDI} = (\text{Health} \times \text{Education} \times \text{Income})^{1/3}$$

It is a **geometric mean**, not a simple average.

So I took this value and multiplied it with 100 to make it a percentage

Recovering Pre-Disaster GDP Growth:

Documentation for recovery_years Feature

Overview

The recovery_years column measures **how long it takes a country to recover to its pre-disaster GDP growth level** after a natural disaster.

We use:

- **DII_v6.csv** → Disaster dataset
- **RSS_v3.csv** → Full annual macro dataset (includes GDP per capita growth)

Only **disaster-year rows** get a recovery_years value.

All non-disaster years remain **NULL**.

Conceptual Definition

For a disaster in **country = ISO, year = Y**:

1. Determine the **pre-disaster GDP growth baseline** using:
 - $Y - 1$
 - $Y - 2$
 - $Y - 3$
2. Compute the **first year after the disaster ($Y+1 \rightarrow 2024$)** where GDP growth returns to or exceeds that baseline:

$$GDPgrowth(ISO, T) \geq baseline$$

3. The number of years needed for recovery is:

$$recovery_years = T - Y$$

4. If the country **never recovers**, use the fallback:

$$recovery_years = 2024 - Y$$

Step-By-Step Logic

1 Compute Pre-Disaster Baseline (3-year window)

We compute the average GDP per capita growth rate over:

- Year-1
- Year-2
- Year-3

Formula:

$$baseline = mean(GDPgrowth[Y - 1], GDPgrowth[Y - 2], GDPgrowth[Y - 3])$$

Rules:

- Missing values are ignored
 - If **all three years missing**, baseline = **NaN**
 - If baseline is NaN → recovery_years stays **NaN**
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2 Search for Recovery (Y+1 → 2024)

Starting at Y+1, check each future year:

“Does GDP growth reach or exceed the baseline?”

If yes, the recovery year is:

$$recovery_years = T - Y$$

Example:

- Disaster in 2005
 - Recovery threshold reached in 2007
→ recovery_years = 2
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3 If No Recovery Occurs

If no year from Y+1 to 2024 reaches the baseline:

$$recovery_years = 2024 - Y$$

This represents the **maximum recovery horizon** given available data.

Edge Case Handling

Missing GDP values

Skipped — they do not reset the counter.

Disaster at dataset boundary

If Y-1, Y-2, Y-3 are outside available data → NaN.

Multiple disasters same year

All disasters for the same ISO/Y receive the **same recovery_years**.

Non-disaster rows

Always **NULL**.

Dataset Flow

Input 1: RSS_v3.csv

Full macroeconomic time series — one row per ISO per year

Contains GDP per capita growth

Input 2: DII_v6.csv

Disaster events — sparse dataset

Contains ISO + Year of disaster

Output: Final merged dataset

RSS_v3 enriched with new column:

recovery_years

- Value only for disaster years
 - Null for all other years
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Final Formula Summary

$$baseline = mean(\{GDP_{Y-1}, GDP_{Y-2}, GDP_{Y-3}\})$$

$$recovery_years = \begin{cases} T - Y & \text{if } \exists T \in [Y + 1..2024] \text{ s.t. } GDP_T \geq baseline \\ 2024 - Y & \text{otherwise} \end{cases}$$